Running app: https://mini-marketplace-se.web.app

Team: Team 01 - Mini Marketplace

Members:

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Scrum Master: @MinhQuangQu - Nguyen Quang Minh (Sprint 5)

Repository: https://github.com/SE-FDA-NEU/G1_AI66A_SE

Release v1.0: https://github.com/SE-FDA-NEU/G1_AI66A_SE/releases/tag/v1.0

Latest green CI run on main: https://github.com/SE-FDA-NEU/G1_AI66A_SE/actions/runs/43567135851

Test accounts: see the first screen of README.md.

Milestone 5 - Deployment, Operations and Final Handoff

Team 01 - Mini Marketplace
Evidence snapshot: 14 November 2026
Reviewed main commit: f826537 (merge of PR #247)
Status: PR #247 merged; main CI green and frontend deployed by CI; release v1.0 published on f826537; branch-protection screenshot outstanding; sprint points are author-approved estimates.

The submission block follows the Sprint 5 brief. Member names, student IDs and the Sprint 5 Scrum Master follow the information supplied by the author. Sprint 4/5 result fields were estimated by the assistant with the author's permission (see 5.1). The linked green CI run is the run on the reviewed main commit f826537, which also deployed the frontend. The team should replace the estimates with its own accepted-work records if they differ.

This single document combines the submission information, product entry points, seven-section operations draft, weekly issue/PR evidence, sprint summary and final-demo plan. Repository documentation includes docs/operations.md, the README first-screen URL/accounts/P0 entries and a Sprint 5 section with the five-sprint table in docs/sprint-log.md; these were merged through PR #247. The course submission is Team01_M5.pdf, containing the required short block; MILESTONE5.pdf is the full supporting report.

The reviewed activity covers 8-14 November, with candidate Sprint 5 issues from 2-14 November. Formal sprint assignment must be confirmed. Closed issues, merged PRs, passing targeted tests, hosted acceptance and a released version are recorded separately. No interviews, user testing or human approvals are invented.

1. Product, test accounts and P0 stories

Mini Marketplace helps Buyers browse products and place orders, while Sellers create listings and manage fulfillment. The intended public interface requires no installation. All six P0 journeys must start from links or buttons on the first page and work on laptop and phone.

1.1. Disposable test accounts

Role	Email	Password	Use
Buyer	user@example.com	password	Cart and checkout
Buyer	demo.buyer@marketplace.local	password	Separate-account ownership checks
Seller	seller@example.com	password	Create listing
Seller	demo.seller@marketplace.local	password	Seeded seller-order demonstration
Admin	admin@example.com	password	Disposable administration demonstration

These are repository-documented sample accounts, not real-user credentials. Verify they exist on the selected demo database before publishing this list. The Admin account also appears in the public development seed. Use disposable data only; anyone who knows these credentials can access exposed demo roles. Do not use these passwords or publish real secrets for a private deployment.

1.2. P0 stories and where to start

Story	Click path from the first screen	Role
US01 Browse catalog	Open running app -> Products	Guest
US02 Product detail	Products -> product card	Guest
US03 Shopping cart	Log in as Buyer -> Add to Cart -> Cart	Buyer
US04 Place order	Cart -> Checkout -> confirm -> confirmation	Buyer
US06 Create listing	Log in as Seller -> Create product	Seller
US08 Manage orders	Log in as demo Seller -> Seller orders -> status action	Seller

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2. Operations

2.1. How to run and release

Running app: https://mini-marketplace-se.web.app

The app uses Firebase Hosting for static HTML/CSS/JavaScript and a Cloudflare Quick Tunnel to FastAPI and host-local SQLite. Follow the repository's deployment guide and SQLite guide. Preserve DATABASE_URL and the private JWT signing secret across restarts. Initialize explicitly; do not reset or reseed an active demo as a startup step.

Release flow: pull request, non-author review, merge into main, CI green, then the Deploy Firebase Hosting job exports the frontend (MARKETPLACE_API_ORIGIN variable, FIREBASE_SERVICE_ACCOUNT secret) and deploys it to the live channel. This sequence ran on main commit f826537 (run 43567135851): tests passed and the deploy job succeeded. The live root and /products returned 200, and the deployed backend-config.js pointed at the tunnel that answered /health with 200. Backend CORS must allow the exact Hosting origin. A changed tunnel URL requires updating the variable and redeploying. Manual deployment (python scripts/export_frontend.py, firebase deploy --only hosting) remains available as a fallback.

The backend is not deployed by CI: it runs on the demo host (python -m src.main with .env.demo, persistent SQLite) behind cloudflared. Tag v1.0 on f826537, the commit whose frontend is live. Roll back frontend assets to a previous Firebase Hosting release and restart the

matching verified backend commit. Alternatively, revert the faulty commit through a reviewed PR and let CI deploy again. Do not downgrade a database schema blindly; stop writes and restore a verified snapshot to a new path only when necessary.

2.2. The CI pipeline

Step on main	Check
Detect stack	Select Python and optional npm jobs
Install	Python 3.12, requirements, pytest/coverage/Ruff/Playwright
Lint	Ruff; currently warning-only
Test	Pytest, coverage output and JUnit report
Page scripts	11 JavaScript DOM/network harnesses on main; not a real-browser walkthrough
Optional Node job	npm installation, lint/test/build if a manifest exists
Secrets	Credential patterns, tracked environment files and database dumps
Summary	Publish pytest totals even after failure

The latest main run (commit f826537) took about 12 minutes, passed 1,802 tests and deployed the frontend: 43567135851. The previous main run, 43063328416, had failed with 16 failures and 18 errors. Main's active ruleset requires one PR approval; it lists no required status checks. Configure those checks and capture a genuine settings screenshot. No such screenshot is included here.

2.3. Monitoring

Read application/Uvicorn logs in the backend process and tunnel logs in the cloudflared process; the backend is not running inside Firebase Hosting. Startup, shutdown, access requests and database exceptions are logged. Passwords, tokens, addresses and request bodies must never be logged. This is a policy to verify: existing exception logging may expose database parameters. Keep DEBUG=false, restrict log access and audit redaction.

The /health probe is served by the backend, so it is reached through the tunnel, not Firebase Hosting (which returns 404 for /health). It checks database connectivity with SELECT 1 and returns 200 when connected and 503 when disconnected (changed in PR #247; the tunnel returned 200 on 14 November). The 503 branch is covered by tests, not by a live outage. Neither proves all P0 flows work. Proposed alerts: 5xx rate above 1% over five minutes and p95

response time above two seconds over five minutes. Check every minute, notify the demo operator, inspect logs and rerun a read-only smoke check. Alerting is not yet verified.

2.4. Known risks and technical debt

Item	Layer	What could happen	Plan
Required checks not enforced	tests	A red change could be merged	#245: require the CI checks in the main ruleset
Temporary tunnel and local host	infrastructure	Host shutdown or a new tunnel URL breaks the app	#245: assign an operator; update MARKETPLACE_API_ORIGIN and redeploy on every tunnel change
/health not on the Hosting URL	infrastructure	A probe of the public URL cannot see backend state	Accepted until stable hosting exists; probe the tunnel /health
Local SQLite and uploaded files	data	Host loss or mismatched backups loses records/images	#245: coordinate snapshots and verify restore
Google account linking incomplete	code	Existing password users cannot sign in with Google	#245, follow-up to #243: authenticated account linking
Sprint 4/5 points missing	docs	The velocity table is incomplete	#245: Scrum Master fills verified points
Exception logs may expose values	code	Private information leaks into operational logs	#245: audit/redact database errors before public demo

2.5. Maintenance plan

Area	What to do	Proposed owner	How often
Bugs	Reproduce, prioritize and assign reports	@leducminh290506-eng	Daily through grading
Dependency updates	Review updates and rerun regression checks	@teddywristh	Weekly and before release

Area	What to do	Proposed owner	How often
Data backups	Snapshot SQLite/uploads; restore separately	@MinhQuangQu, demo operator	Daily; restore before release
Monitoring	Review availability, errors and latency	@MinhQuangQu, demo operator	Every minute automatically; daily review
Documentation	Check URLs, accounts and actual commands	@minhnm162	Every release/configuration change

2.6. Lessons learned

1. Catalog changes left seed expectations and contracts inconsistent, and main CI failed. PR #247 repaired the fixtures and main is green again. Next time, review fixtures, documentation and full CI together before merging; targeted passes are insufficient.
2. Buyer/Seller navigation exposed irrelevant actions (#220, #221, #225). Reviewed PRs separated the controls. Next time, specify a role-by-action matrix and cover both visibility and server authorization from the start.
3. Tunnel-dependent image URLs can retain an obsolete hostname. PR #242 documents this limitation. Next time, choose a stable API/image origin before releasing seeded or uploaded content to a shared environment.

2.7. Release and handoff checks

Issue #245 stays open until the team accepts the handoff; PR #247 is merged. The author identifies @MinhQuangQu as Sprint 5 Scrum Master. Remaining: capture the branch-protection screenshot, confirm the estimated sprint points and the grading-end date, and run the live desktop/mobile P0 walkthrough. Keep the host and tunnel available through the confirmed grading date. The evidence inventory and finishing checklist are included below.

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3. Sprint 5 issue and PR evidence

Issue	Assigned work	Owner(s)	GitHub state	PR / evidence and remaining limitation
#220	Hide Cart/Add to Cart for Seller	@minhnm162	Closed	#231 merged; approval recorded; Python check successful
#221	Hide Seller-only actions for Buyer	@teddywristh	Closed	#232 merged; changes requested then approval; Python check failed
#222	Checkout UI and delivery/voucher input	@leducminh290506-eng	Closed	#230 merged; changes requested then approval; Python check successful
#223	Show every item in order detail	@MinhQuangQu	Closed	#236 merged; changes requested then approval; Python check failed
#224	Seller shipping-status progression	@leducminh290506-eng	Closed	No dedicated merged PR established in this week's inventory; trace implementation and verify transitions live
#225	Remove Buyer purchase-review controls for Seller	@minhnm162	Closed	#235 merged; approval; Python check failed
#226	Product categories and images	@teddywristh	Closed	#233 merged; changes requested then approval; Python check failed
#227	Unread notification badge and mark-all-read	@minhnm162	Closed	#237 merged; approval; Python check failed

Issue	Assigned work	Owner(s)	GitHub state	PR / evidence and remaining limitation
#228	Marketplace home navigation link	@leducminh290506-eng	Closed	#234 merged; approval; Python check failed
#229	UI/UX redesign	All four members	Closed	No single dedicated merged PR established here; link concrete before/after changes before claiming completion
#238	Firebase frontend Hosting setup	@teddywristh	Closed	#241 merged; approval; Python check failed
#239	Persistent SQLite and backup/restore	@MinhQuangQu	Closed	#242 merged; approval; some restart/restore results reported; full AC not verified
#240	HTTPS tunnel and frontend/API integration	@MinhQuangQu	Closed	Same PR #242; remaining tunnel restart, checkout, roles, upload and recovery checks explicitly listed
#243	Google sign-in	@MinhQuangQu	Closed	#244 merged; approval; real-account sign-in and safe linking still open
#245	Sprint 5 wrap-up and handoff	@MinhQuangQu	Open	#247 merged (approved by @leducminh290506-eng); Python check green; CI deploy succeeded. #246 was

Issue	Assigned work	Owner(s)	GitHub state	PR / evidence and remaining limitation
				closed and recreated as #247

All 11 merged PRs have a recorded non-author approval. This does not prove each approval applies to its final head. Only #230 and #231 have successful Python checks at the inspected heads; the other nine do not. Therefore the minimum of four merged, reviewed, green PRs is **not established** by this inventory. Fourteen closed issues and eleven merged PRs are activity totals, not velocity or proof that every acceptance criterion passed.

3.1. Per-member contribution evidence

Member	Merged authored PR example	Review given example	Closed assigned issue example
@teddywristh	#232, #233, #241	Approved #235	#221, #226, #238
@MinhQuangQu	#236, #242, #244	Requested changes then approved #230	#223, #239, #240, #243
@minhnm162	#231, #235, #237	Requested changes then approved #232	#220, #225, #227
@leducminh290506-eng	#230, #234	Requested changes then approved #233	#222, #224, #228

These examples satisfy the activity categories within the reviewed dates; confirm formal Sprint 5 assignment before marking the sprint requirement done.

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4. Release readiness and deployment alignment

Requirement in the brief	Evidence on 14 November	Required next step
Public HTTPS app	https://mini-marketplace-se.web.app/ and /products returned 200 on 14 November (HTTP check, not a private-window or phone test)	Verify in private windows on laptop and phone
All six P0 stories by clicks	README provides entry points; API smoke checks through the live origin: product	Run US01/02/03/04/06/08 by clicking from the first page

Requirement in the brief	Evidence on 14 November	Required next step
	list, and Buyer/Seller/demo Seller logins succeeded	
Real persistent data	Demo SQLite holds 53 products and the five accounts; PR #242 reports data surviving restart and restore	Independently verify create/reload/backend restart/tunnel restart/init/restore
Public app /health 200/503	Tunnel /health returned 200 with database: connected; the 503 branch is test-covered; Hosting /health returns 404	Show the tunnel /health in the demo; accept or fix the Hosting limitation
Backend available	Backend and a Cloudflare Quick Tunnel were started on 14 November; the deployed frontend points at it	Keep the host awake; redeploy if the tunnel URL changes
Green main CI	Run 43567135851 on f826537: 1,802 passed, deploy job succeeded	None
CI-gated deployment	Deploy Firebase Hosting ran after green checks on main	Re-verify on the next merge
Protected main	Active organization ruleset requires one approval; no required-status-check rule listed	Require relevant CI checks and attach a genuine settings screenshot
Four reviewed green merged PRs	Of the eleven earlier PRs only #230 and #231 have green Python heads; #247 is merged, approved and green	Three qualifying PRs established (#230, #231, #247); one more is needed
Per-member contributions	All four members have examples above	Confirm sprint membership and record evidence
SM wrap-up closed	#245 open; @MinhQuangQu is the author-identified SM and assignee	Close only after handoff acceptance
Sprint log and five-sprint table	Sprint 4/5 filled with author-approved estimates	Replace with accepted-work records if different
README first screen	URL, badge, accounts and P0 links merged	None

Requirement in the brief	Evidence on 14 November	Required next step
v1.0 matches deployed SHA	Release v1.0 targets f826537, the deployed commit	None
Availability through grades	Date set to 14 November 2026 as a placeholder; operator @MinhQuangQu	Confirm the real date

The earlier green run 42495790436 (commit 2d12635, 8 November) is superseded by run 43567135851. The failures of run 43063328416 (stale seed counts, Google error-code documentation, the image-field label and product-detail fixture foreign keys) were fixed in PR #247; the last of these needed the fixture to create its own seller with fixed ids.

Public endpoints checked on 14 November: https://mini-marketplace-se.web.app/ (200), /products (200) and /health (404, no backend route on Hosting).

4.1. Alignment with the deployment guide

Checked against main f826537. The deployment architecture agrees with this pack: Firebase serves the allow-listed static export; FastAPI and SQLite remain on the backend host; Cloudflare exposes only the application API. Firestore's optional seed is not the application's active data source.

Deployment guide / implementation	Corresponding report statement	Result
.firebaserc uses mini-marketplace-se; firebase.json serves public/	Firebase frontend Hosting	Matches
export_frontend.py copies 14 selected pages, static assets and a 404; excludes uploads	Backend, database, environment and uploads remain outside Hosting export	Matches
MARKETPLACE_API_ORIGIN generates static/backend-config.js; CORS permits exact frontend origins	One public API origin; re-export/redeploy when tunnel changes	Matches; explicit steps added to operations draft
Manual hosting:channel:deploy preview / deploy --only hosting	CI deploy job exists and ran on f826537	Matches; manual deploy remains a fallback
Probe /health on the local backend and public tunnel	Backend connectivity differs from app-URL health required by M5	Clarified; Firebase has no /health rewrite

Deployment guide / implementation	Corresponding report statement	Result
SQLite backup API and restore to a new path; uploads backed up separately	Persistent database and coordinated database/image backup	Matches; successful full restore still needs evidence
Temporary tunnel and absolute uploaded-image URLs	Availability and stale image-hostname risks	Matches

The deployment guide's default live URL is now the submission URL. The backend and tunnel are not deployed by CI and depend on the demo host. Sprint 4/5 result fields remain blank as requested; release publication and the real-browser P0 walkthrough remain pending.

Local validation on 14 November: full pytest on the branch before the last fixture fix gave 1,784 passed and 18 errors (the product-detail fixture); the fixed module passes, and CI on main then passed 1,802 tests. The 11 page-script harnesses and tests/backend_runtime_ui.mjs pass locally. A real-browser hosted walkthrough has not been done.

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5. Sprint summary and velocity

5.1. Sprint 5 - delivery and operations

Goal: deliver the six P0 journeys at a public HTTPS URL with persistent demo data, CI-gated releases, health checks, operational ownership and a verifiable v1.0 release. Candidate dates are 2-15 November from the preceding sprint cadence; exact sprint dates remain unconfirmed. Sprint 5 Scrum Master: @MinhQuangQu, as identified by the author.

Measure	Value	Why
Committed story points	46	**Estimate** (author-approved): 15 issues sized by assistant, no recorded points
Completed story points	43	**Estimate**: 14 closed issues; #245 stays open
Velocity	43	Equals estimated completed points
Recent issue activity	14 closed, 1 open	Fifteen candidate-window issues; #245 remains open

Measure	Value	Why
Recent PR activity	12 merged (including #247), 0 open	#246 was closed and recreated as #247

Sprint	Committed points	Completed points	Velocity	Evidence/limit
1	39	39	39	Historical sprint log; specification work
2	33	33	33	Historical log includes technical-task points
3	35	35	35	Historical log excludes 11 supporting hours
4	14	14	14	Committed 14 from the planning log; completed is an estimate: #17, #19, #21, #22 are all closed
5	46	43	43	**Estimate**, author-approved; sizes: bugs 2, tasks/features 1-5, #245 3

Preserve these historical conventions and explain the mixed work types; do not recalculate velocity from the whole Project board or convert estimated hours to story points. This task did not independently verify the historical totals. No attendance, retrospective meeting, interview or user test is invented.

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6. Final demo - 12 minutes plus 5 minutes of questions

Time	Show	Preparation and evidence
0:00-1:00	Problem/persona: Buyers purchase; Sellers list and fulfill products. Name six P0 stories.	US01 catalog, US02 detail, US03 cart, US04 checkout, US06 create listing, US08 seller orders
1:00-6:00	All six journeys on final URL in a private window, using clicks; create and reload a record; reject one server-side business rule	Prepare disposable accounts/data. Buyer: catalog -> detail -> cart -> checkout; Seller: create listing -> orders -> status. Show stock-limit rejection and recovery. Include two documented M3-feedback before/after pairs.
6:00-8:00	M4 tests, normal/error/boundary cases, one red test then undo	Open latest green test step and docs/testing.md; rehearse a local stock-rule mutation on an isolated copy. Restore immediately; never mutate the public demo.
8:00-10:00	Main Actions, required checks/approval, deployment, rollback, public /health; defend one design choice	Explain keeping the existing structured FastAPI/SQLite monolith versus hosted PostgreSQL; trigger to reconsider: durable hosting, concurrency or multiple backend instances. Do not claim stable hosting has been achieved.
10:00-12:00	One evidence-based lesson and one risk with its issue/owner	Use role-specific UI fixes or failing CI; discuss temporary tunnel availability and #245 follow-up
Questions: 5 minutes	Individual explanations and another team's question	Each member opens an authored PR/test/screen and explains rules, failures and tradeoffs

Existing M3 screenshots are under docs/images/ui/ and responsive evidence under docs/images/responsive/. Select before/after pairs that correspond to documented M3 feedback in docs/evidence/milestone3.md; do not relabel every weekly UI change as M3 feedback. No new hosted screenshots were captured here.

6.1. Two UI changes to prepare for the live demonstration

Change documented in M3 evidence	Before	Recorded improvement	Remaining demo evidence
Confirmation receipt readability	Product displayed as a database ID and unformatted amount	Product name/code and formatted price; see docs/evidence/milestone3.md, section 5	Select the authentic old screenshot and capture the matching current hosted screen
Seller-order recovery/navigation	Missing navigation link, technical network text and list disappearing after a failed change	Seller-orders link, login redirect, actionable feedback and retained list; same M3 evidence section	Select the authentic old screenshot and capture the matching current hosted error state

These are documented M3 improvements, not independently confirmed post-M3-feedback changes. The team must connect the selected examples to actual feedback and show the old and new screenshots during the demo. Do not label a reconstruction as an old screenshot or claim a new user evaluation.

6.2. Boundary and red-test rehearsal

Use the cart stock rule in docs/testing.md: with stock 10, quantities 9 and 10 are permitted, while 11 is refused without changing the stored cart. Rehearse on an isolated local database/copy: run the relevant cart test, introduce one stock guard defect, show that test fail, undo the defect and show it pass again. This is a rehearsal plan; no mutation demonstration is claimed to have been run here.

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7. Finishing checklist

1. Fill the final URL, confirmed sprint dates/points and grading-end date; keep the supplied Scrum Master username consistent.

2. Repair CI in the M5 branch; obtain teammate review of code, docs and deployment gate.
3. Require green checks on main and capture the real settings screenshot.
4. Merge reviewed work, verify green main CI and record the full deployed frontend/backend SHA pair.
5. Verify public /health 200/503, all six P0 flows, roles, images and backend failure recovery on laptop and phone.
6. Verify persistence across backend/tunnel restarts and restore of database plus uploaded files; assign backup/monitoring owners.
7. Resolve incomplete #243/#239/#240 acceptance checks or record explicit follow-up; close #245 only after handoff acceptance.
8. Done: docs/operations.md, README and sprint-log edits were merged through PR #247 after a non-author review.
9. After deploy succeeds, tag the exact verified main commit v1.0 and publish a release with three factual lines. Do not tag an arbitrary latest commit.
10. Fill the remaining URL/points fields, update release and green-run evidence, then regenerate and inspect Team01_M5.pdf; submit that short PDF, not the full supporting report.

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8. Sources, evidence limits and author declaration

- Sprint 5 wrap-up #245
- M5 handoff PR #247, merged as f826537
- Main README
- Main CI workflow
- Main ruleset
- Project board

Deployment references:

- Deployment guide
- Persistent demo SQLite
- M4 testing documentation
- M3 UI evidence

Public availability was checked by HTTP requests, not a private-window browser or physical phone. Earlier hosted browser and persistence results are attributed to their PR authors. Proposed monitoring thresholds, maintenance owners and release procedures require team confirmation.

The grading-end date remains unspecified in the supplied brief. Historical sprint totals are reproduced from the existing log and were not independently audited.

AI assistance: OpenAI Codex assisted with reading the supplied Sprint 5 brief, reviewing repository and GitHub evidence, drafting and consolidating this report, and executing the documented local checks. Team members must review the final report and explain their own implementation and evidence.

- ☐ The team has reviewed this report and corrected remaining blank fields.
- ☐ The author has named contributors and can explain the submitted changes.
- ☐ A teammate who did not author the work has submitted a substantive review.
- ☐ The deployed version, green CI, public health (tunnel) and v1.0 release are verified; Hosting /health returns 404.

On 14 November the assistant set the MARKETPLACE_API_ORIGIN variable, started the backend and tunnel, and recreated the PR; it did not merge, review or create the release. Local work for #239 was discarded at the user's request; the report references the merged implementation in PR #242.

9. Continuation handoff

Field or evidence	Current state	Who must provide or verify it
Final running HTTPS URL	https://mini-marketplace-se.web.app	Demo operator; test all P0 journeys on phone and laptop
Sprint 4 committed / completed / velocity	14 / 14 / 14 (completed estimated)	Scrum Master to confirm
Sprint 5 committed / completed / velocity	46 / 43 / 43 (estimated)	Scrum Master to confirm
Branch protection screenshot	Not captured	Repository maintainer; show approval and required green checks
Latest green main run and deployed SHA	Run 43567135851, commit f826537, green and deployed	None; re-verify after any new merge
Release v1.0	Published on f826537	None
Public /health 200 and 503	Tunnel returns 200; 503 test-covered; Hosting returns 404	Backend/deployment owner

Field or evidence	Current state	Who must provide or verify it
Grading-end date and availability owner	14 November 2026 (placeholder chosen by the author) / @MinhQuangQu	Team/lecturer to confirm the real date

Continue from main f826537. Preserve supplied member IDs and @MinhQuangQu as the author-identified Scrum Master. Do not calculate velocity from issue counts or estimated hours. Keep author declarations unchecked until the human author confirms them. Do not tag, change repository protection or replace the public deployment solely to make this document appear complete.